

School of Electrical, Electronic and Mechanical Engineering

EEME30001 - Embedded Computing Systems

Coursework Description - 2026/27

A Real-Time Embedded Control System

Marking Criteria

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Coursework Marking Criteria

1 Introduction

The four coursework criteria have been designed to align directly with the four unit learning outcomes.

This marking scheme is a discipline-specific implementation of the University of Bristol Level 6 marking framework. The descriptors within each criterion have been adapted from the University Level 6 Marking Criteria to reflect the specific context of Embedded Computing Systems while maintaining alignment with the University's expectations regarding student attainment at Level 6.

The coursework is assessed against four criteria, each corresponding directly to one of the unit learning outcomes. For each criterion, the marker will determine the most appropriate attainment band based on the evidence presented in the submission.

The examples provided in this document are illustrative only. They are not requirements and should not be interpreted as a checklist.

2 Criterion Mapping Summary

The coursework criteria have been designed to align directly with the unit learning outcomes while also maintaining alignment with the University of Bristol Level 6 marking framework and the Engineering Council AHEP4 learning outcomes.

The University Level 6 criteria reflected within this assessment are:

- Content Knowledge
- Critical Approach
- Explanation and Interpretation
- Problem Solving
- Specialist Skills and Techniques

This assessment also contributes evidence towards individual student attainment of the following AHEP4 outcomes:

- **C1**: Engineering Knowledge
- **C3**: Experimental Methods
- **C6**: Systems Integration
- **C12**: Practical Application

Table 1 shows how each coursework criterion maps onto the selected University Level 6 marking criteria. Table 2 shows how each criterion contributes evidence towards the relevant AHEP4 outcomes.

These mappings are provided for transparency and guidance. They are intended to explain the educational purpose of each criterion and how the assessment has been designed. They should not be interpreted as additional assessment criteria, nor should they be used as a marking checklist.

The assessment should continue to be understood primarily through the four published coursework criteria and their associated learning outcomes.

Criterion	Content Knowledge	Explanation & Interpretation	Critical Approach	Problem Solving	Specialist Skills & Techniques
Criterion A	✓	✓			
Criterion B				✓	✓
Criterion C		✓		✓	
Criterion D			✓	✓	✓

Table 1: Mapping between assessment criteria and the selected University Level 6 marking criteria.

Criterion	C1	C3	C6	C12
Criterion A	✓		✓	
Criterion B		✓	✓	✓
Criterion C			✓	✓
Criterion D		✓	✓	

Table 2: Mapping between assessment criteria and AHEP4 outcomes.

3 Final Mark Calculation

The purpose of these criteria is to support classification of attainment against each learning outcome. They are not intended to be used as a score sheet or mark-calculation tool.

> The final mark is determined through academic judgement, taking account of the submission as a whole, and is not calculated by combining criterion scores, attainment bands or weighted averages.

The University's 0–20 marking scale (the "21-point scale")[1]is an ordinal scale rather than a percentage scale. Consequently, criterion classifications should not be interpreted as numerical scores and should not be combined using arithmetic averaging or weighting rules. The purpose of the classifications is to support academic judgement rather than replace it.

Your final feedback will include one sentence summarising how the bands were used to derive the final mark.

Different patterns of attainment across the four criteria may reasonably lead to different overall marks. Two submissions showing similar criterion profiles may therefore receive different final marks where the overall quality of attainment, technical achievement or engineering judgement differs.

The exact mapping between the 21-point scale (0–20 scale) and the unrestricted 0–100 point scale can be found in Sec. 8.

[1]University of Bristol Assessment Regulations and Marking Criteria, Sec. 20.13.

4 Criterion A – Learning Outcome 1

Demonstrate a comprehensive understanding of embedded computer operational principles, illustrating how they can be effectively integrated into systems for control and monitoring.

Alignment with University Level 6 Marking Criteria

- Content Knowledge
- Explanation and Interpretation

AHEP4 Alignment

- C1
- C6

Attainment Bands

0–5

Content Knowledge: Gaps in knowledge with only superficial understanding of embedded-system principles. Significant inaccuracies are present and key concepts are absent, confused or incorrect.

Explanation and Interpretation: Provides weak and/or poorly constructed explanation of system behaviour, implementation decisions or the interaction between system components.

Illustrative Examples: Major misunderstandings; unable to explain how key parts of the system operate.

6–8

Content Knowledge: Demonstrates sufficient understanding of core embedded-system principles. Some awareness of more specialised system operation is evident, although significant gaps remain.

Explanation and Interpretation: Demonstrates some ability to explain the implemented system and its operation, although explanations may lack detail, consistency or technical depth.

Illustrative Examples: Can explain what the system does and how major components interact.

9–11

Content Knowledge: Demonstrates sound and broad understanding of embedded-system principles and their application within the implemented system.

Explanation and Interpretation: Provides generally clear and consistent explanations of system operation, component interaction and implementation choices.

Illustrative Examples: Correct explanations of architecture, timing behaviour, state-machine operation and software structure.

12–14

Content Knowledge: Demonstrates systematic and coherent understanding of embedded-system principles and their application within the implemented system.

Explanation and Interpretation: Explains system operation thoroughly, clearly, coherently and consistently. Demonstrates the ability to justify implementation decisions and interpret observed system behaviour.

Illustrative Examples: Clear discussion of design decisions, behaviour and implementation choices. May explain why particular approaches were selected.

15–17

Content Knowledge: Demonstrates excellent knowledge and understanding of embedded system concepts. Understanding extends beyond implementation details and shows awareness of broader engineering considerations.

Explanation and Interpretation: Provides thorough, perceptive, insightful and well-justified explanations and interpretations of system behaviour, implementation decisions and observed results. Demonstrates strong awareness of limitations, assumptions and consequences of design choices.

Illustrative Examples: May discuss the advantages and disadvantages of polling and interrupt-driven approaches, or evaluate alternative implementation choices.

18–20

Content Knowledge: Demonstrates exceptional, sophisticated, highly-detailed knowledge and understanding of embedded-system principles and their application.

Explanation and Interpretation: Provides exceptional technical insight and highly developed interpretation of system behaviour, observations and implementation decisions. Demonstrates the ability to critically evaluate alternative approaches and communicate reasoning with authority.

Illustrative Examples: Demonstrates sophisticated perception and mature understanding comparable to professional engineering practice. May critically evaluate architectural choices, resource constraints or alternative system designs.

5 Criterion B – Learning Outcome 2

Integrate hardware and software components into a functional embedded system, conducting tests and debugging to ensure proper functionality.

Alignment with University Level 6 Marking Criteria

- Specialist Skills and Techniques
- Problem Solving

AHEP4 Alignment

- C3
- C6
- C12

Attainment Bands

0–5

Problem Solving: Demonstrates little or no evidence of an ability to identify aspects of complex embedded computing systems problems or apply appropriate approaches to address them.

Specialist Skills and Techniques: Has not demonstrated sufficient evidence of effective application of specialist skills and techniques in the implementation and integration of embedded computing systems.

Illustrative Examples: Major functionality missing or system does not operate reliably.

6–8

Problem Solving: Demonstrates limited ability to identify the key aspects of complex embedded computing systems problems or select appropriate approaches to address them.

Specialist Skills and Techniques: Demonstrates evidence of developing specialist skills and techniques in the implementation and integration of embedded computing systems.

Illustrative Examples: Main states implemented and system broadly functions.

9–11

Problem Solving: Shows ability to identify aspects of complex embedded computing systems problems and apply appropriate approaches to address them.

Specialist Skills and Techniques: Demonstrates consistent informed application of specialist skills and techniques in the implementation and integration of embedded computing systems.

Illustrative Examples: Functional state machine, configurable behaviour and working output.

12–14

Problem Solving: Demonstrates evidence of an ability to identify key aspects of complex embedded computing systems problems and apply appropriate approaches to achieve effective solutions.

Specialist Skills and Techniques: Demonstrates consistent capable and effective application of specialist skills and techniques in the implementation, testing and integration of embedded computing systems.

Illustrative Examples: Reliable implementation, clear software structure and successful integration of system functionality.

15–17

Problem Solving: Demonstrates well-developed problem-solving skills, showing the ability to identify key aspects of complex embedded computing systems problems and apply appropriate approaches to address them effectively.

Specialist Skills and Techniques: Demonstrates consistent evidence of accomplished and innovative application of specialist skills and techniques in the implementation, testing and integration of embedded computing systems, including appropriate use of advanced techniques where justified.

Illustrative Examples: Robust implementation, effective integration of multiple elements and strong technical execution.

18–20

Problem Solving: Demonstrates highly developed problem-solving skills, showing confidence, initiative and creativity in identifying key aspects of complex embedded computing systems problems and applying appropriate approaches to solve them.

Specialist Skills and Techniques: Demonstrates consistent accomplished and exceptionally innovative application of specialist skills and techniques in the implementation, testing and integration of embedded computing systems, including highly effective use of advanced techniques where appropriate.

Illustrative Examples: Highly polished and technically accomplished implementation demonstrating exceptional integration of hardware and software components.

6 Criterion C – Learning Outcome 3

> *Implement appropriate interfacing techniques and communication methods during the development of embedded systems. Apply effective testing and debugging methods.*

For this assessment, "interfacing techniques and communication methods" refers to software interfaces and communication between system components.

Alignment with University Level 6 Marking Criteria

- Problem Solving
- Explanation and Interpretation

AHEP4 Alignment

- C6
- C12

Attainment Bands

0–5

Problem Solving: Demonstrates little or no evidence of effective testing, debugging or interfacing activity. Limited evidence exists that communication mechanisms or software interfaces were considered during development.

Explanation and Interpretation: Weak and/or poorly constructed explanation of testing activities, debugging approaches, system behaviour or interface design decisions.

Illustrative Examples: Little evidence that system behaviour has been verified.

6–8

Problem Solving: Demonstrates limited ability to identify key aspects of testing, debugging and validation of a complex system. Some evidence of communication or interfacing mechanisms is present, although implementation or application remains basic.

Explanation and Interpretation: Some ability to explain and discuss testing outcomes, debugging activities or the behaviour of communication and interfacing mechanisms.

Illustrative Examples: Basic testing and limited discussion of outcomes.

9–11

Problem Solving: Demonstrates ability to identify aspects of a complex system and to undertake appropriate testing, debugging and validation. Shows ability to apply suitable communication or interfacing approaches within the implemented system.

Explanation and Interpretation: Provides evidence supporting correct system operation. Provides clear and consistent explanation of testing outcomes.

Illustrative Examples: Testing demonstrates that key functionality is working. May employ interfaces between software modules or simple communication mechanisms to support system operation.

12–14

Problem Solving: Demonstrates ability to identify key aspects of a complex system and to undertake effective testing, debugging and validation. Applies appropriate communication and interfacing approaches to support system functionality, reliability and maintainability.

Explanation and Interpretation: Evidence is well presented and supports evaluation of system performance. Demonstrates the ability to thoroughly interpret observations and explain complex interface or communication decisions.

Illustrative Examples: Structured testing and discussion of findings. May employ clear software interfaces between conceptually distinct components of the system, for example state-machine, PWM and diagnostics modules.

15–17

Problem Solving: Demonstrates well-developed testing, debugging and validation skills. Shows initiative and autonomy in selecting communication and interfacing approaches and applying them appropriately to solve implementation challenges.

Explanation and Interpretation: Provides thorough, perceptive and convincing evaluation of system behaviour, testing outcomes and communication/interface design decisions.

Illustrative Examples: Strong evidence-based evaluation and reflection. May justify interface choices between software modules, communication between execution contexts, synchronisation approaches or the use of diagnostic interfaces during development.

18–20

Problem Solving: Demonstrates exceptional testing, debugging and validation capability. Shows sophisticated application of communication and interfacing approaches to support reliable and maintainable embedded-system operation.

Explanation and Interpretation: Provides authoritative interpretation of observations, measurements and system behaviour. Demonstrates exceptional technical insight into the relationship between testing, debugging, communication mechanisms and system design.

Illustrative Examples: Sophisticated validation and analysis of system behaviour. May demonstrate effective use of DMA-mediated communication, double-buffering, synchronisation mechanisms, critical sections, race-condition avoidance or well-designed system–human communication interfaces.

7 Criterion D – Learning Outcome 4

Design complex embedded systems, taking into consideration real-time requirements and performance parameters in resource- and input/output-constrained environments.

Alignment with University Level 6 Marking Criteria

- Critical Approach
- Problem Solving
- Specialist Skills and Techniques

AHEP4 Alignment

- C3
- C6

Attainment Bands

0–5

Critical Awareness: Demonstrates little or no critical awareness of embedded systems, their limitations or relevant design constraints.

Problem Solving: Demonstrates little or no evidence of an ability to identify aspects of complex problems or apply appropriate embedded-systems approaches to address them.

Specialist Skills and Techniques: Demonstrates little or no effective application of specialist skills and techniques in the development of embedded computing systems.

Illustrative Examples: Design decisions are absent, inappropriate or provide insufficient evidence of embedded-systems design capability.

6–8

Critical Awareness: Demonstrates little critical awareness of embedded systems and their design constraints.

Problem Solving: Demonstrates limited ability to identify aspects of complex problems or select appropriate embedded-systems approaches to address them.

Specialist Skills and Techniques: Demonstrates limited application of specialist skills and techniques in the implementation of embedded computing systems.

Illustrative Examples: Implements a state-machine architecture and demonstrates some awareness of system behaviour and operating requirements.

9–11

Critical Awareness: Demonstrates limited critical awareness of embedded systems and the consequences of design decisions.

Problem Solving: Demonstrates evidence of an ability to identify aspects of complex problems and apply appropriate embedded-systems techniques to address them.

Specialist Skills and Techniques: Demonstrates consistent informed application of specialist skills and techniques in the implementation and testing of embedded computing systems.

Illustrative Examples: Design choices are generally appropriate to the requirements of the system and demonstrate awareness of real-time operation and implementation constraints.

12–14

Critical Awareness: Demonstrates clear critical awareness of embedded systems, design constraints and the consequences of engineering decisions.

Problem Solving: Demonstrates evidence of an ability to identify key aspects of complex problems and apply appropriate embedded-systems approaches to achieve effective solutions.

Specialist Skills and Techniques: Demonstrates consistent, capable and effective application of specialist skills and techniques in the implementation, testing and evaluation of embedded computing systems.

Illustrative Examples: May separate functionality into modules such as state-machine, PWM and diagnostics components to improve maintainability and clarity. Demonstrates coherent architectural organisation and effective decomposition of functionality.

15–17

Critical Awareness: Demonstrates excellent critical awareness of embedded systems, their limitations and the implications of engineering decisions.

Problem Solving: Demonstrates well-developed skills in identifying key aspects of complex problems and applying appropriate embedded-systems approaches to address them effectively.

Specialist Skills and Techniques: Demonstrates consistent accomplished and innovative application of specialist skills and techniques in the development, testing and evaluation of embedded computing systems.

Illustrative Examples: May demonstrate appropriate handling of trade-offs such as simplicity versus responsiveness, polling versus interrupt-driven operation, or processor utilisation versus implementation complexity. May demonstrate well-defined software structure and clear separation of responsibilities.

18–20

Critical Awareness: Demonstrates exceptional critical awareness of embedded systems, their limitations and the implications of design decisions in constrained environments.

Problem Solving: Demonstrates exceptional ability to identify key aspects of complex problems and apply appropriate embedded-systems approaches to solve them, showing initiative and creativity.

Specialist Skills and Techniques: Demonstrates consistent accomplished and exceptionally innovative application of specialist skills and techniques in the design, implementation and evaluation of embedded computing systems.

Illustrative Examples: May demonstrate sophisticated architectural organisation, highly effective use of embedded-systems techniques, or particularly strong consideration of constraints, performance and maintainability in the overall design.

8 University Mark Conversion Scale

The University permits assessment using either the 0–100 marking scale or the 0–20 marking scale. Marks awarded using the 0–20 scale are converted to fixed points on the 0–100 scale in accordance with University regulations.[2]

Table 3: Relationship between the 0–20 and 0–100 marking scales

0–20 Scale	**Equivalent 0–100 Mark**
20	100
19	94
18	83
17	78
16	75
15	72
14	68
13	65
12	62
11	58
10	55
9	52
8	48
7	45
6	42
5	35
4	29
3	22
2	15
1	7
0	0

The 0–20 scale is a non-linear ordinal scale. A mark awarded on the 0–20 scale must not be interpreted as a percentage and cannot be converted by simple multiplication. The conversion relationship shown in Table 3 is defined by University regulations and is used wherever marks on the two scales must be combined, compared or converted.

[2] Taught Code of Practice Section 20: Marking Criteria and Scales